

C. PRANAV SHARMA

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Education

Amrita Vishwa Vidyapeetham

B.Tech in Computer Science and Engineering | CGPA: 8.23/10

Coimbatore, India

Aug 2023 – Apr 2027

Relevant Coursework: Embedded Systems, Cyber Security, Distributed Systems, Machine Learning, Probability & Statistics

Technical Skills

Languages: Java, Python, JavaScript, C

Frameworks & Backend: Node.js, Express.js, FastAPI, RESTful API Design

Databases: MongoDB, Firebase

Frontend: React.js, TailwindCSS

Systems & Tools: Linux, Docker, Git, Wireshark, Keil, Cisco Packet Tracer

Core CS: Data Structures & Algorithms, Operating Systems, DBMS, OOP, ML

Projects

AegisBERT: Zero-Day Threat Intel Engine | *PyTorch, FastAPI, Docker, Streamlit* | [GitHub](#) April 2026
Fine-tuned ModernBERT on 150,000+ custom-mined CVEs, across 13 threat categories for automated vulnerability classification.
Engineered an **automated NVD ingestion pipeline** using hybrid labeling (CWE mapping + regex heuristics), eliminating manual annotation.
Optimized inference using **Flash Attention and mean pooling**, achieving < 200 ms latency with support for **8192-token context**.
Deployed a **Docker-based microservices architecture** integrating an asynchronous FastAPI backend with a Streamlit dashboard for real-time analytics and visualization.

SentinelScore: Production MLOps Credit Risk Engine | *XGBoost, Optuna, MLflow, DVC* | [GitHub](#) March 2026
Built an end-to-end credit risk scoring system on **300K+ loan records** with **120+ engineered features**, improving **PR-AUC by 18%** using Bayesian optimization (Optuna).
Designed a **fully reproducible ML pipeline using DVC** with stage-level dependency tracking, reducing redundant re-computation by **60%**.
Implemented **privacy-preserving data pipelines** using SHA-256 + HMAC anonymization to ensure complete PII protection before training.
Integrated **MLflow experiment tracking and model registry**, logging **50+ runs** with metrics, parameters, and feature importance artifacts.
Deployed a **Dockerized FastAPI inference service** with dynamic feature reconstruction and **F1-optimized thresholds**, achieving **less than 120ms latency**.

AgriWise – AI-Driven Agricultural Marketplace | *React, Node.js, MongoDB, Socket.io* | [GitHub](#) February 2026
Collaborated in a **4-member team** to build a full-stack **AI-powered agricultural marketplace** enabling farmers to sell crops with quality-based dynamic pricing.
Developed a **multimodal grading engine** using Google Gemini 2.5 Flash and engineered a **3-tier market data pipeline** (Gov API → CSV Cache → Synthetic Grid) ensuring **100% price availability**.
Built a **real-time event-driven backend** using Express and Socket.io with **scalable MongoDB schemas** and REST APIs to support auctions, order lifecycle, and concurrent operations.
Deployed the system using **Vercel and Render** and implemented **CI/CD pipelines** for automated builds and reliable updates.

Secure Sealed Bid E-Tendering System | *Python, Flask, Cryptography, MongoDB* | [GitHub](#) February 2026
Designed a secure backend system implementing **role-based access control (RBAC)** with **TOTP-based multi-factor authentication** for controlled tender participation.
Developed a complete **public-key cryptographic workflow using RSA** to ensure sealed bid submission, digital signatures, and tamper-proof verification.
Implemented secure data pipelines using **hashing, Base64 encoding, and encrypted storage**, ensuring confidentiality and integrity of sensitive bid data.
Built RESTful APIs for full tender lifecycle management, supporting **concurrent bid submissions and controlled reveal mechanisms** during evaluation.

Embedded Defence Radar System | *Embedded C, STM32, PWM, I2C* | [GitHub](#) June 2025
Developed a real-time radar simulation system on **STM32 ARM Cortex-M4** using bare-metal Embedded C with interrupt-driven architecture.
Implemented **ultrasonic pulse-echo sensing** and PWM-controlled servo scanning for continuous semi-circular object detection.

Built low-level **I2C drivers** to interface with OLED display, enabling real-time radar visualization with dynamic detection indicators.